

Sandra Tang

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EDUCATION

Cornell University | College of Engineering

Ithaca, NY

B.S. in Computer Science

Expected May 2027

Relevant Coursework: Computer System Organization and Programming, Analysis of Algorithms, Object-Oriented Programming & Data Structures, Discrete Structures, Functional Programming

TECHINICAL EXPERIENCE

Imaging Systems Software Engineer

Ithaca, NY

Cornell Unmanned Air Systems

February 2025 – Present

- Developed systems software coordinating onboard cameras, flight telemetry, and ground systems for a UAV imaging platform operating under real-time constraints.
- Implemented a time-aligned, real-time data pipeline to synchronize image capture with telemetry streams across distributed components.
- Integrated Pixhawk flight telemetry via MAVLink/MAVProxy by designing a timestamp-based buffering and nearest-sample matching system to align images with GPS and attitude data.
- Wrote asynchronous, fault-tolerant workflows to handle latency, packet loss, and intermittent connectivity during flight.
- Owned end-to-end development and testing of the imaging pipeline, from design and implementation through live flight operations and debugging.

Backend Engineering Intern

Ithaca, NY

Data Legion AI

August 2025 – November 2025

- Designed and built backend services to ingest, process, rank, and serve large document collections through modular, multi-stage pipelines.
- Implemented APIs and storage workflows supporting multi-stage processing and downstream consumers, emphasizing correctness and reliability.
- Containerized services to support deterministic builds and configuration management across environments.

Freelance Web Developer

Boston, MA

Hao Shi Guang Restaurant

June 2025 – Present

- Built and deployed a full-stack web application with live users, owning development, deployment, and maintenance.

Software Engineer

Ithaca, NY

Cornell Engineering World Health

Nov 2024 – May 2025

- Developed backend logic supporting structured data delivery for a curriculum platform used in resource-constrained environments.

PERSONAL PROJECTS

StudyCentral App – Backend Lead

Ithaca, NY

Cornell AppDev Hack Challenge

December 2024

- Designed and implemented REST APIs and containerized backend services to support repeatable development, testing, and deployment workflows.

Handwriting Recognition Web App

Boston, MA

Tufts University

July 2023

- Developed an end-to-end handwriting recognition system combining numerical data processing, model inference, and API-based serving.

Video Classification Model

Boston, MA

Sensorium

June 2023 – August 2023

- Fine-tuned a Video Vision Transformer (ViViT) for high-dimensional video data, focusing on performance and resource constraints using qLoRA optimization.

TECHNICAL SKILLS

Programming Languages: Rust, C++, C, Python, Java, Go, JavaScript, SQL, HTML, CSS, TypeScript, OCaml

Systems & Infrastructure: Real-time and asynchronous pipelines, fault-tolerant systems, distributed services, hardware/software interfaces

Tools & Libraries: Linux, Docker, Git, FastAPI, REST APIs, AWS, React.js, Node.js, OpenCV, NumPy, Pandas, TensorFlow, Spring Boot, MySQL, Azure

ML: PyTorch, Hugging Face Transformers, Computer Vision